

On the origin of isotropic steady state structural superplastic deformation

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ABSTRACT

Many models are available to explain isotropic steady state structural superplastic deformation. One such proposal suggests that grain boundary sliding (GBS) that develops to a mesoscopic scale is the rate controlling process in the above mentioned range in different classes of materials with different crystal structures. The equations derived accurately describe the experimental results pertaining to many superplastic materials of different classes, crystal structures and grain sizes tested at different temperatures and strain rates. In this paper using existing experimental data available in the open literature the isotropic steady state structural superplastic deformation is predicted accurately by first evaluating the model constants using some of the existing experimental data. Using these model constants, the isotropic steady state structural superplastic deformation of some other superplastic material conditions not used for the analysis is predicted once again to high accuracy. As required by a sound theory, it is pointed out that some predictions of the physical description have already been verified experimentally by earlier findings.

1. Introduction

A sound physical theory evolves as follows: 1. Finding of some major experimental observations concerning a physical phenomenon → 2. Formation of postulates/assumptions as to its origin → 3. Proposal of a physical theory → 4. Verification of the assumptions and validation of the theory using experimental results → 5. New predictions of the model → 6. Their verification/validation by experimental findings.

The “rediscovery” by Backofen et al. (1964) [1] of the phenomenon of structural superplastic deformation constituted a major landmark. Materials of different classes of a certain microstructure, obtained by suitable thermomechanical processing, attain the superplastic state in which the flow stress is highly grain-size-sensitive and the application of a small tensile stress gives rise to extreme elongations of several hundred/a few thousand percent in tensile specimens tested at temperatures in the range of $0.25 T_m < T < T_m$, with the best results displayed in many cases when T , the temperature of deformation on the absolute scale, lies in the region $0.4 T_m - 0.8 T_m$, where T_m is the melting temperature of the material on the absolute scale.

The spectacular observation regarding superplastic deformation is the extreme elongation of several hundred percent in a tensile test [2], in contrast to what is seen after tensile deformation dominated by dislocation motion by glide, climb and related accommodation processes, all

of which limit plastic deformation even in the best ductile materials to usually less than $\sim 40\text{--}60\%$. This unusual behaviour of superplastic materials resembles that of hot/molten polymers, plasticine, tar, silly putty, etc. Notwithstanding this observation, many superplastic materials, e.g., metallic and ceramic materials, are crystalline. It is interesting that this phenomenon has been reported in metallic materials, ceramics, metal-as well as ceramic-composites, intermetallic compounds, nano-structured materials, metallic glasses and geological materials of FCC, BCC, HCP and other crystal structures of lower symmetry [3–7]. In tensile tests on conventional materials, grain elongation in the tensile direction is approximately proportional to the external, macroscopic specimen elongation. In contrast, even after extreme superplastic deformation the grain size remains nearly equiaxed and at no stage the grain size aspect ratio is greater than $\sim 1.2\text{--}1.4$. When the initial microstructure is textured and anisotropic, the deformation is also anisotropic until the texture is randomised in an initial region of transient flow and the grain shape becomes equiaxed [7–9].

A cylindrical tensile specimen made of a material with an anisotropic microstructure became elliptical on superplastic deformation. But if the microstructure was isotropic, the cross-section remained truly circular [3]. Transient flow region is extremely limited in strain if the starting microstructure is isotropic. This transient flow region is also described as the region of quasi-viscous flow during which a material starts from rest

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and gradually reaches the isotropic, steady state superplastic deformation with a characteristic delay time to reach the steady state. In this region of deformation, in the absence of grain growth (due to static and (high-homologous-temperature-dependent) dynamic grain growth), the flow stress is independent of strain and can be expressed as a single valued function of grain size, strain rate and the absolute temperature of deformation [6].

As mentioned above, even after several hundred percent tensile elongation, the grain shape remains nearly equiaxed and any texture present in the starting specimen gets randomised by superplastic deformation. No new preferred orientations develop, and the lattice dislocations present in the starting material due to prior history, e.g., a cold-rolled sheet specimen, get annihilated by emission into the grains during the initial transient region of flow, which then traverse the grain to the opposite boundaries and get delocalised there. The grain shape becomes “bulbous” in appearance and the grain boundaries are “smoothed”, all of these are indicative of dominant grain boundary sliding and near-random grain rotation as a concomitant of a grain boundary sliding controlled process with unbalanced grain boundary shear stresses. There is no evidence for intra-granular dislocation motion. Whatever dislocation motion inside the grains that has been reported has been interpreted by most of the authors as due to the emission of lattice dislocations generated when grain boundary sliding results in the required misfit to emit a lattice dislocation of the right Burgers vector on to the concerned grain boundary plane. Such a dislocation then traverses the grain to be absorbed at the opposite boundary. There is some evidence (e.g., the presence of denuded zones) for limited diffusion, which is regarded in the model as a faster accommodation process compared with grain boundary sliding. High-angle grain boundaries are the most important structural elements where the rate controlling grain boundary sliding process is concentrated. When the grain boundaries in a material are only of the low-angle type, superplastic behaviour is lost (for a summary of experimental results in this regard, see, [10]).

A thorough assessment of the experimental results by several authors has led to the conclusion that isotropic steady state superplastic deformation is the result of a mesoscopic grain-boundary-sliding-controlled process accommodated by highly localised, atom shuffling events parallel (grain boundary sliding) and perpendicular (grain boundary migration) to the grain boundary [10–19]. This conclusion contradicts the dominant view held till early 1980s that grain boundary sliding is an inherently fast process and that at no stage it can be rate controlling [20]. This debate got settled finally in favour of near-total-strain resulting from grain boundary sliding after some very clear high resolution scanning transmission electron microscopic studies culminating in the two papers of Ikuhara and coworkers [18,19], and Wang et al. [21]; for a recent assessment, see Padmanabhan and Divinski [22].

It is well known that a mismatch between the rate of grain boundary sliding and that of the accommodation process will lead to grain boundary cavitation. This problem becomes more severe if the mechanical properties of the constituent phases are significantly different, the applied stress is greater than the shear rupture stress of the interface between the two phases of differing mechanical properties, local stress concentration etc. This observation suggested - and subsequent experiments verified it - that by applying a “back pressure” the cavitation tendency can be suppressed and this knowledge led to the development of “back pressure forming”, which enhanced the flexibility and scope of defect-free superplastic forming. This effect of back pressure revealed that the hydrostatic component of the stress system influences the yield behaviour of superplastic materials and that in principle anisotropic yield criteria like those of Mohr-Coulomb, Gurson-Rice-Tetelman, etc., should describe superplastic yielding more accurately than the von Mises criterion, which assumes that the hydrostatic component of the stress system has no influence on the yield behaviour. But the other two criteria mentioned above require the assumption of empirical fitting constants, unlike the von Mises criterion, which requires no empirical/

fitting constants. It has already been shown [17,23,24] that the difference between the predictions of these hydrostatic-stress-dependent criteria and the Mises criterion is negligible and so it is safe to use the Mises criterion of yielding for understanding superplastic deformation (as a good approximation) as it does not require any empirical fitting constants.

2. Physical model

Many theoretical explanations are available for isotropic, steady state structural superplastic deformation [7]. The model discussed in this paper suggests that there is a unique underlying physical mechanism that is responsible for isotropic, steady state structural superplastic deformation that cuts across different classes of materials, crystal structures of the constituent phases etc. Such an approach minimises the number of assumptions and will adhere to the proposition that “to the same natural effects we must, as far as possible, assign the same causes” [25].

The near-random grain rotations and texture randomization during superplastic deformation can be understood only if the unbalanced shear stresses at the boundaries arising from grain size and shape irregularities and orientation differences between the individual boundaries to the external stress survive during deformation. This observation supports the idea of grain boundary sliding control. Diffusion control will not introduce any change in the texture and grain rotations will not be present. Dislocation control will lead to typical hot working/cold working textures. Such textures are never observed after isotropic, steady state structural superplastic deformation. Invoking a second rate controlling mechanism in addition to grain boundary sliding is unnecessary because such a step will add to the number of constants necessary to explain the experimental results. At the level of atomistics, “atom shuffling”, indicative of grain boundary sliding (GBS; parallel to the boundary) and grain boundary migration (GBM; perpendicular to the boundary), has been observed in experiments and MD simulation [18, 19,21].

Further, superplastic deformation in different classes of materials can be understood as the physical picture suggests the development of GBS to a mesoscopic scale (defined to be of the order of a grain diameter or more) [9,12,14–16,23,26–31]. If the barriers to boundary sliding at triple junctions and other grain boundary obstacles are accommodated by faster accommodation processes like dislocation emission from sliding boundaries (in a barrier-free manner or during a thermal event [12,32]), localised diffusion and/or grain rotation, GBS can develop to a mesoscopic scale and form plane interfaces. By interconnection, such plane interfaces can form a 3D-network of deforming high-angle grain/interphase boundaries and lead to superplastic flow. Experimental evidence for mesoscopic boundary sliding/plane interface formation has been reported in many systems, see, for example, [13,33–36]. The length of the plane interfaces formed will be a function of grain size and temperature. The plane interface formation process will require work to be done by the external stress and give rise to a long-range threshold stress (i.e., required for rearranging matter in the vicinity of grain boundaries) which should be subtracted from the applied stress to know the effective stress that drives the deformation process. Predictions exist that the long-range threshold stress increases with decreasing grain size and/or decreasing temperature. These predictions have been experimentally verified/validated [3,23]. The accommodation process that removes the grain boundary obstacles will depend on the nature and strength of the grain boundary obstacles, nature of interatomic bonds, chemical composition of the material, phases present and experimental conditions like strain rate, grain size and temperature. Such a description can easily explain how superplastic deformation is equally dominant in materials of FCC, BCC, HCP or other crystal structures in metallic materials, ceramics, bulk metallic glasses, composites, intermetallic compounds, nanostructured materials, geological materials and ice/ice-mixture. It could also explain why the grain shape remains

near-equiaxed even after extreme specimen elongation.

For quantification, the basic sliding unit in a high-angle boundary is assumed to be of an oblate spheroid shape of 5 atom diameters on the boundary plane and 2.5 atom diameters ($\approx$ average grain boundary width [37]) in height measured from the centre of the oblate spheroid in a direction perpendicular to the boundary plane [12,15]. Such a choice of dimensions allows superplastic deformation in crystalline and non-crystalline (e.g., bulk metallic glasses) materials to be described equally well, in addition to being consistent with the experimental results [12,30,38]. In real materials there is a grain size distribution and grain shape irregularities. In this model these are replaced by equal volume grains of rhombic dodecahedral shape (closest to the shape of real crystals [39]). When the shape of the basic boundary sliding unit deviates from an oblate spheroid shape and is irregular (the real case), it is replaced by an equal volume oblate spheroid and correction factor $\alpha_f \approx 1$.

Superplastic deformation is a high homologous temperature process. Therefore, the grain size increases with time, temperature and strain. For the general case an equation for grain growth should be used as a secondary kinetic equation. But static as well as dynamic grain growth in materials is still an unsolved problem. Therefore, empirical equations have been used to describe grain growth. The steady state stress-strain rate relationship in a material in a given state (annealed or cold/warm worked) commences soon after the initial transient deformation is completed. As superplastic materials are often grain-coarsening resistant (of micro-duplex or poly-phase microstructures), the mean grain size at the beginning of the tests can be used as the mean grain size of the deforming material as reasonable approximation to achieve a very good fit with the experimental data (please see below). Finally, as mentioned earlier, superplastic materials are mostly micro-duplex or poly-phase. In this analysis the grain boundaries of different kinds are replaced by a “representative” grain boundary. The activation energy/free energy of activation for boundary sliding, Q depends on G the shear modulus, V_0 the volume of the basic oblate spheroid shearing unit, γ_0 the unit shear strain that results from a unit boundary sliding event, and the threshold shear stress necessary for GBS to develop to a mesoscopic scale τ_0 , which depends on γ_0 , the specific grain boundary free energy γ_B , and the number of boundaries N_a , that align to form a plane interface. N_a will depend on grain size, temperature, the average behaviour of the different types of boundaries, i.e., it depends on the atom species, phases present in the material, and how they are arranged in the boundaries due to the differences in boundary misorientation and the nature of the chemical bonds. It should be noted the τ_0 can be expressed as a function of the other variables and so the independent model constants are γ_0 , G , γ_B , N_a , and a , the average grain size exponent that accounts for the grain size and shape distributions, as well as static and dynamic grain growth. Of these, calculation of G and γ_B from first principles, e.g., electron theory of metals, constitutes separate problems that are beyond superplastic deformation and is applicable to all materials – both superplastic and non-superplastic. The other three of constants γ_0 , N_a , and a are unique to the phenomenon of isotropic steady state superplastic deformation. How the fundamental studies to determine these three constants are presently being pursued are summarised, for example, in the works [8,9,12,14–16,23,27].

3. Methodology used to determine the model constants

Let there be N materials conditions (materials of different processing history) of many classes of materials such as metals and alloys, inter-metallic compounds, ceramics, composites, bulk metallic glasses, geological materials and nanostructured materials that exhibit isotropic steady state structural superplastic deformation under certain thermal and microstructural conditions.

At constant microstructure, the real activation energy for isotropic steady state structural superplastic deformation is calculated using Eq.

(1),

$$\dot{\epsilon} = \left(\frac{A_1}{T}\right) \sigma^n \exp\left(-Q/kT\right); \text{ if it is taken that } \nu = (10^{13} \text{ s}^{-1}) \quad (1)$$

A_1 is a grain-size-dependent (but temperature and stress independent) constant that has absorbed the other constants of the model and $\dot{\epsilon}$ is the tensile strain rate, which is equal to $\left(\frac{\dot{\gamma}}{\sqrt{3}}\right)$, with $\dot{\gamma}$ the shear strain rate. (It is already shown that the difference in the results is negligible if $\left(\frac{kT}{h}\right)$ is used instead of 10^{13} s^{-1} as the thermal vibration frequency [23].) Then, Q for every material condition can be calculated from the slope of the $\ln \dot{\epsilon}$ against $\left(\frac{1}{T}\right)$, keeping σ^n constant. Here, n is the stress exponent, intrinsically a function of temperature and grain size, obtained as the instantaneous slope of the isothermal, isostructural $\ln \sigma - \ln \dot{\epsilon}$ plot, σ the tensile stress ($= (\sqrt{3} \tau)$), and τ the shear stress. Then, the *material-condition compensated average value of the activation energy for isotropic steady state superplastic deformation*, Q_{av} , can be expressed as,

$$Q_{av} = \sum_{i=1}^i \sum_{j=1}^j \left[\frac{Q_{ij}}{T_{mi}} - a - \frac{b}{T_{ij}} \right]^2 \quad (2)$$

According to the model, the activation energy for the rate controlling process is expressed as [12,14],

$$Q = \frac{1}{2} (\beta_1 \gamma_0^2 + \beta_2 \epsilon_0^2) G V_0 \quad (3)$$

Therefore, from every value of Q , G for every material condition can be calculated using the following facts.

For the oblate spheroid shape assumed for the basic unit of GB sliding, $\beta_1 = 0.944 \left(\frac{(1.590-p)}{(1-p)} \right)$; $\beta_2 = \frac{4(1+p)}{9(1-p)}$, where p is the Poisson ratio. (When this value is unknown, it is assumed to be equal to 0.33) [40]. V_0 is the volume of the oblate spheroid – the volume of the basic boundary sliding unit - i.e., $V_0 = \frac{2}{3} \pi W^3$, where W is 2.5 times the atomic diameter a_0 , where a_0 is the atomic diameter of the major constituent element of the material. γ_0 is the mean shear strain associated with a basic/unit boundary sliding event, $\epsilon_0 = \left(\frac{\gamma_0}{\sqrt{3}} \right)$ and G is the shear modulus. Estimates of the best fit values of G , γ_0 and ϵ_0 for all material systems can be obtained by using the method of least squares expressed below as Eq. (4).

$$\sum \left[Q - \frac{1}{2} (\beta_1 \gamma_0^2 + \beta_2 \epsilon_0^2) G V_0 \right]^2 \quad (4)$$

Please note that there will be N values of γ_0 and N values of G . Using the N values of γ_0 , calculate $\gamma_{0i \text{ mean}}$ as a function of temperature at constant grain size, $\gamma_{0i \text{ mean}} = \left(a_{\text{mean}} + \frac{b_{\text{mean}}}{T_i} \right)$ and similarly estimate $\epsilon_{0i \text{ mean}} = \left(a_{\text{mean}} + \frac{b_{\text{mean}}}{T_i} \right)$; recall $\left(\epsilon_0 = \left(\frac{\gamma_0}{\sqrt{3}} \right) \right)$. Also, using the N values of G , calculate the mean value of G/T_m as $\left(\frac{G_i}{T_{mi}} \right)_{\text{mean}} = \left(a_{\text{mean}} + \frac{b_{\text{mean}}}{T_i} \right)$, i.e. represent G as a function of T_m and T at constant grain size for all the material conditions. (The absence of sufficient experimental data prevents grain size also to be included as a part of the developed equations.)

The superplastic shear strain rate, given in Eq. (5), is [17],

$$\dot{\gamma} = \frac{2W\gamma_0\nu}{d} \sin h\left(\frac{(\tau - \tau_0)\gamma_0 V_0}{kT}\right) \exp\left(\frac{-Q}{kT}\right); \tau < \tau_0 \quad (5)$$

Here, d is the grain size, τ and $\dot{\gamma}$ are the externally applied shear stress and shear strain rate respectively and can be obtained from experimental tensile stress-strain rate data considered up to the optimal point (the point of inflection on the isothermal, isostructural sigmoidal curve)

at which the strain-rate sensitivity index is maximum. τ_0 is the shear stress needed to induce grain boundary sliding on a mesoscopic scale, which leads to the formation of a mesoscopic plane interface containing N_a contiguous grain boundaries that align to form the boundary sliding plane interface that shears on a mesoscopic scale. Using Eq. (5), τ_0 for every superplastic material condition is estimated by the method of least squares as there are several stress-strain rate combinations for every material condition and temperature and the mesoscopic threshold stress variation is represented as $\left(\frac{\tau_0}{T_m}\right)_{mean} = \left(a_{mean} + \frac{b_{mean}}{T_i}\right)$ for all the superplastic material conditions. (Grain size is constant.)

From the model it also follows that the threshold shear stress, τ_0 necessary for grain boundary sliding to develop to a mesoscopic scale, can be estimated using Eq. (6), where γ_B is the specific grain boundary free energy, N_a is the number of grain boundaries that are aligned to form the mesoscopic plane interface and a is a grain-size exponent that depends on the grain shape and grain size distribution and degree of grain growth [14].

$$\tau_0 = (2G\gamma_B)^{0.5} \left[\frac{0.6204}{N_a^{0.5}d} \right]^a \quad (6)$$

Estimate the best fit values of γ_B , N_a and a for every material condition at the different grain sizes and temperatures using the method of least squares with the τ_0 values available at each temperature and grain size of every material and express them as a function of grain size and temperature. Based on the already available experimental results, it can be stated that γ_B must lie between 0.30 and 1.7 and its value should be varied in this range with an increment of 0.001. N_a must lie between 2 and 30 and its value should be varied in this range with an increment of 0.1. When the grains are irregular in shape and there is also a grain size distribution, as in real materials, the value of a lies between 0.1 and 0.5 and this value should be varied in this range with an increment of 0.001 [14].

Prediction of strain rates: Using the expressions for Q , G , γ_B , γ_0 , ε_0 , τ_0 , N_a and a , the present authors estimated the shear strain rate using Eq. (5) for every material condition at all grain sizes and temperatures reported and compared the results with the experimental shear strain rates. The prediction accuracy was evaluated using the *root mean square error* (RMSE) and the *tolerance* value, which is defined as the ratio of the predicted to the experimentally observed shear strain rate. By convention the larger of the two values is taken on the numerator. Also, the prediction accuracy was tested further using other material conditions of different grain sizes and temperatures which were not used for the above analysis.

4. Results and discussion

4.1. Analysis to obtain the mean model constants and predicting the strain rates of deformation for all the material conditions

Using the procedures described above, the expressions for the material properties/parameters Q , G , γ_B , γ_0 , ε_0 , τ_0 , N_a and a were obtained for different classes of superplastic material conditions that exhibit isotropic, steady state structural superplastic deformation. In all, experimental data pertaining to **146 material conditions** were analysed and the results are tabulated in Table 1. The activation energy, Q , was calculated for 41 materials and using the method of least squares as shown in Eq. (2), the expression for the mean value of Q/T_m is estimated as shown in Eq. (7).

$$\frac{Q}{T_m} \left(\frac{J}{Mol.K} \right) = 135.6 - \frac{5195.08}{T} \quad (7)$$

Using the estimated activation energy for different material conditions and using Eq. (4) the expressions for the mean value of G/T_m , γ_0 and ε_0 ($\varepsilon_0 = \left(\frac{\dot{\gamma}}{\sqrt{3}}\right)$) were estimated as shown below in Eqs. (8)–(10).

$$\frac{G}{T_m} \left(\frac{MPa/K}{K} \right) = 26.45 - \frac{928.98}{T} \quad (8)$$

$$\gamma_0 = 0.0827 + \frac{1.3422}{T} \quad (9)$$

$$\varepsilon_0 = 0.0408 + \frac{0.0117}{T} \quad (10)$$

Using the above relationships, τ_0 for every material condition was estimated by the method of least squares using the shear strain rate Eq. (5), with a constraint that τ_0 should be less than the minimum experimentally reported value in every campaign. The mean value of τ_0/T_m is estimated as a function of temperature as shown in Eq. (11). The variations of Q/T_m , G/T_m , γ_0 and τ_0/T_m with temperature are given in Fig. 1.

$$\frac{\tau_0}{T_m} \left(\frac{MPa/K}{K} \right) = 4.593 \times 10^{-3} + \frac{0.4511}{T} \quad (11)$$

The expressions for γ_B , N_a and a are obtained using the method of least squares from the τ_0 values available for every temperature and grain size. The expression

$\left[\tau_0 - (2G\gamma_B)^{0.5} \left[\frac{0.6204}{N_a^{0.5}d} \right]^a \right]^2$ is minimised to obtain the best combination of a , γ_B and N_a values for each combination of grain size and temperature for every material. The expression for γ_B , N_a and a as functions of temperature and grain size are given in Eqs. (12)–(14).

$$\gamma_B = 1.075 - \frac{19.962}{T} \quad (12)$$

$$a = 0.185 + 0.0108.d + 1.37 \times 10^{-4}.T - 3.7 \times 10^{-4}.d^2 - 5 \times 10^{-8}.T^2 + 2.37 \times 10^{-8}.d.T \quad (13)$$

$$N_a = 9.625 + 0.568.d + 1.11 \times 10^{-2}.T + 1.04 \times 10^{-2}.d^2 - 3.85 \times 10^{-6}.T^2 - 8.85 \times 10^{-4}.d.T \quad (14)$$

To establish the accuracy of the analysis employed, a prediction of strain rates is made for all the material conditions subjected to different thermo-mechanical conditions. In all, experimental data pertaining to 467 material conditions were used and the tolerance values were tabulated. The comparison of predicted shear strain rates with the experimental values and the corresponding tolerance values for one material from every class of material is given in Table 2 and full comparison of predictions for all material conditions is given in supplementary material, Table S.1. Around 10 % of the predicted strain rates were negative and hence omitted from Table 2 and Table S.1. The RMSE value for the analysis is 0.0435 and the percentage of predictions with a tolerance value greater than 10 is 26 %. The prediction is very accurate in the range of strain rates 10^{-6} to $10^{-2} s^{-1}$. Large deviations come from the larger strain rates. For example, for the material, La₅₅Al₂₅Ni₂₀ [70], the prediction is very good at the lower strain rates, but as the strain rate increases, the prediction accuracy suffers. From Table 2 and Table S.1, if we remove the predictions with a tolerance values greater than 10 at the extremes of strain rates (less than $10^{-6} s^{-1}$ and greater than $10^{-2} s^{-1}$), the percentage of predictions with a tolerance value greater than 10 is reduced to 15 %.

4.2. Predictions concerning systems not used for the analysis

Using the same expressions for Q , G , γ_B , γ_0 , ε_0 , τ_0 , N_a and a , the superplastic response of the following materials Zn22Al [79], Al6Cu0.4Zr [80], Pb–Sn eutectic [81], β -SiC_w/ZK60 Composite [82], pure Ice, pure Ice +10%CaF₂ [83], Mg 3Zn 0.5 Y 1.5 Zr [84], Olivine, Fo90 [85] and Si₂O₁₃ [86], which were not included in the analysis to obtain the predictive equations, were predicted. The comparison of predicted shear

Table 1

Estimated values of model parameters for different material conditions.

Sl. No.	System	Grain size (μm)	T (K)	T_{hom}	γ_0	ε_0	Q/T_m (J/Mol K)	G/T_m (MPa/K)	τ_0/T_m (Pa/K)	γ_B (J/m ²)	a -	N_a -
Metals and Alloys												
1	Zn 22Al [41]	2.5	423	0.558	0.059	0.034	77.49	51.38	2902.33	1.29	0.262	14.21
			473	0.624					1675.44	1.11	0.267	14.49
			503	0.664					1187.32	0.91	0.270	14.64
2	Zn 22Al [42]	0.9	453	0.597	0.061	0.035	103.18	49.39	3614.76	1.31	0.248	14.05
			518	0.683					3179.44	1.18	0.254	14.49
			545	0.719					2255.92	0.75	0.256	14.65
3	Al 33Cu 0.4Zr [43]	7.6	713	0.869	0.066	0.038	170.35	41.65	146.16	1.61	0.332	15.76
			753	0.917					109.66	0.84	0.335	15.71
			793	0.966					97.44	0.56	0.338	15.65
4	Al 5.76Mg 0.32Sc 0.3Mn [44]	3	723	0.775	0.089	0.051	135.44	20.29	2154.32	1.57	0.293	15.57
			748	0.802					1886.36	1.58	0.295	15.64
			773	0.829					1532.69	1.48	0.297	15.70
5	Al 5.76Mg 0.32Sc 0.3Mn [45]	1	523	0.561	0.083	0.048	131.86	22.78	13365.50	0.92	0.255	14.53
			573	0.614					9035.41	0.69	0.260	14.83
			623	0.668					7427.64	0.48	0.264	15.12
6	Al 3Mg 0.2Sc [46]	0.2	573	0.614	0.086	0.050	118.56	21.87	5798.52	1.46	0.250	14.77
			623	0.668					4823.12	1.32	0.254	15.09
			673	0.721					3579.84	1.03	0.258	15.39
7	Al 8.9Zn 2.6Mg 0.009Sc [47]	0.7	723	0.775	0.080	0.046	134.19	25.27	2904.58	1.08	0.261	15.67
			493	0.528					26323.70	1.47	0.249	14.29
			523	0.561					18553.02	0.87	0.252	14.49
8	Al 5Mg 0.18Mn 0.2Sc [48]	24	563	0.603	0.090	0.052	141.75	19.87	9217.57	0.84	0.255	14.75
			583	0.625					6623.76	0.71	0.257	14.87
			603	0.646					4544.52	0.73	0.259	14.99
9	Al 17Si 2Fe 2Mg 1Cu 1Ni [49]	1.4	748	0.802	0.089	0.051	116.48	23.53	1350.44	0.87	0.347	19.57
			773	0.829					1264.73	0.97	0.350	19.17
			793	0.85					1243.28	0.97	0.352	18.85
10	Mg 6Zn 0.8Zr [50]	0.7	813	0.871	0.080	0.046	128.70	18.04	675.26	0.77	0.354	18.52
			763	0.818					9024.68	0.88	0.278	15.78
			783	0.839					4855.29	0.93	0.280	15.86
11	Mg 4Y 0.7Zr 0.4Nd [51]	2	793	0.85	0.087	0.050	132.28	15.21	3204.71	0.82	0.280	15.89
			803	0.861					1061.12	0.87	0.281	15.93
			448	0.485					9296.55	1.54	0.245	13.98
12	Mg 5.8Zn 1Y 0.48Zr [52]	17.5	473	0.512	0.080	0.046	128.70	18.04	7456.74	1.65	0.247	14.16
			498	0.539					5660.17	1.02	0.250	14.33
			523	0.566					4523.81	1.35	0.252	14.49
13	Ti 6Al 4V [53]	0.9	597	0.647	0.087	0.050	132.28	15.21	5021.66	1.35	0.273	15.04
			623	0.674					1655.82	1.29	0.275	15.16
			648	0.701					1331.16	1.38	0.277	15.28
14	Cu 2.8Al 1.8Si 0.4Co [54]	7	673	0.728	0.088	0.510	113.57	15.38	1201.29	0.89	0.279	15.39
			673	0.728					3528.13	1.37	0.358	18.11
			723	0.783					1937.25	1.58	0.363	17.62
15	Ti 6Al 4V [53]	0.9	753	0.815	0.116	0.067	124.29	11.56	790.08	1.23	0.367	17.32
			1023	0.529					4066.21	1.14	0.286	16.73
			1073	0.555					3155.75	0.82	0.288	16.84
16	Cu 2.8Al 1.8Si 0.4Co [54]	7	1123	0.581	0.096	0.056	106.59	24.61	2581.50	0.73	0.290	16.94
			1173	0.607					2333.16	0.88	0.291	17.01
			723	0.533					8134.25	1.48	0.328	15.70
17	Ti 43Al [56]	5	773	0.57	0.095	0.055	107.17	25.19	4306.82	0.99	0.332	15.66
			823	0.607					2138.61	1.17	0.336	15.60
			873	0.644					1563.45	1.04	0.340	15.52
18	Ti 48Al [57]	0.9	673	0.497	0.081	0.047	108.91	33.01	6939.53	0.99	0.289	15.41
			723	0.533					4609.12	0.78	0.293	15.57
			773	0.57					3613.60	0.53	0.297	15.70
19	Ti 46.2Al 2.2Cr [58]	0.8	823	0.607	0.095	0.055	107.17	25.19	2396.76	0.69	0.300	15.82
			735	0.652					19591.86	1.34	0.289	15.61
			773	0.686					12928.10	1.26	0.291	15.73
20	IN836 [55]	2.5	815	0.723	0.081	0.047	108.91	33.01	7444.52	0.88	0.294	15.85
			853	0.757					4986.65	1.29	0.296	15.95
			876	0.777					5226.26	0.87	0.298	16.00
Intermetallic Compounds												
17	Ti 43Al [56]	5	1273	0.695	0.144	0.088	154.81	7.23	7394.89	1.45	0.341	15.07
			1323	0.723					3752.03	1.32	0.342	14.91
			1373	0.75					2211.94	1.49	0.343	14.73
18	Ti 48Al [57]	0.9	1163	0.654	0.138	0.080	160.71	7.73	23813.29	1.20	0.291	17.00
			1273	0.716					3458.96	1.02	0.293	17.11
			1373	0.772					1906.65	0.81	0.294	17.12
19	Ti 46.2Al 2.2Cr [58]	0.8	1073	0.599	0.125	0.072	146.66	9.39	22138.44	1.07	0.287	16.88
			1123	0.627					12317.11	0.99	0.288	16.98
			1173	0.655					4042.43	0.77	0.290	17.06
20	Co ₃ Ti [59]	24	1173	0.776	0.096	0.055	180.25	23.78	26514.51	0.86	0.389	12.15
			1223	0.809					21375.63	0.76	0.393	11.18

(continued on next page)

Table 1 (continued)

Sl. No.	System	Grain size (μm)	T (K)	T_{hom}	γ_0	ε_0	Q/T_m (J/Mol K)	G/T_m (MPa/K)	τ_0/T_m (Pa/K)	γ_B (J/m ²)	a -	N_a -
21	Ni ₃ Si [60]	15	1273 1323 1353 1373	0.842 0.882 0.902 0.915	0.092	0.053	147.16	31.01	19259.30 6179.96 3493.29 933.33	1.09 0.87 0.56 0.93	0.397 0.406 0.407 0.408	10.20 10.97 10.59 10.34
Ceramics/Composites												
22	ZrO ₂ [61]	0.07	1273 1323 1373	0.426 0.443 0.46	0.082	0.047	112.41	17.79	13330.00 10314.60 4769.08	1.65 1.40 1.06	0.282 0.282 0.283	17.56 17.62 17.65
23	ZrO ₂ 3Y [62]	0.51	1523 1573 1623 1673 1723	0.51 0.526 0.543 0.56 0.577	0.088	0.051	122.47	15.38	1572.93 1362.08 1258.41 927.01 321.25	0.85 1.10 1.08 0.96 1.03	0.288 0.288 0.287 0.285 0.284	17.30 17.24 17.16 17.07 16.95
24	ZrO ₂ 4Y [63]	0.75	1573 1623 1673 1723	0.526 0.543 0.56 0.577	0.089	0.051	126.12	15.05	592.37 230.96 117.14 56.87	1.57 1.17 1.67 1.17	0.291 0.290 0.289 0.288	17.05 16.96 16.85 16.72
25	Al ₂ O ₃ 30ZrO ₂ 30Al ₁₆ Si ₂ O ₁₃ [64]	0.4	1673 1723 1773	0.723 0.745 0.767	0.067	0.039	144.83	33.26	1919.62 1456.95 1279.69	0.86 1.26 1.19	0.284 0.283 0.281	17.16 17.05 16.92
26	Al ₂ O ₃ 25NiAl ₂ O ₄ 25ZrO ₂ [65]	1.3	1623 1648 1673 1698	0.702 0.713 0.723 0.734	0.064	0.037	153.98	40.41	4690.84 4647.61 4578.44 4517.91	1.35 1.13 1.01 1.28	0.298 0.297 0.297 0.296	16.49 16.43 16.36 16.29
27	6061/ 20 % SiC [66]	0.8	773 823 843	0.59 0.628 0.643	0.071	0.041	114.57	37.50	3954.22 2312.96 1061.03	1.05 1.00 1.19	0.272 0.275 0.276	15.87 16.09 16.17
28	7075/ 20 % SiC [67]	5	753 773 793	0.596 0.612 0.627	0.069	0.040	117.39	39.33	7220.88 4528.89 3452.14	0.97 1.14 1.05	0.314 0.316 0.317	15.62 15.64 15.65
Bulk Metallic Glasses												
29	Zr ₆₅ Al ₁₀ Ni ₁₀ Cu ₁₅ [68]	~10	653 673 693 713	0.587 0.605 0.623 0.641	0.074	0.043	124.71	26.27	13821.94 7239.25 5350.72 4559.31	1.35 1.00 0.80 0.57	0.340 0.342 0.344 0.346	16.22 16.17 16.11 16.05
30	Zr _{52.5} Al ₁₀ Cu ₂₂ Ti _{2.5} Ni ₁₃ [69]	~10	683 693 703	0.621 0.63 0.639	0.075	0.043	133.19	27.60	19572.74 13181.85 6781.81	0.78 0.97 0.49	0.343 0.344 0.345	16.14 16.11 16.08
31	La ₅₅ Al ₂₅ Ni ₂₀ [70]	~10	483 493 503 513	0.687 0.701 0.716 0.73	0.059	0.034	139.26	30.18	14096.72 12119.48 8193.45 6756.79	1.60 1.30 1.27 1.28	0.322 0.323 0.324 0.326	16.57 16.56 16.54 16.53
32	La ₆₀ Al ₂₀ Ni ₁₀ Co ₅ Cu ₅ [71]	~10	460 480 500 520	0.713 0.744 0.775 0.806	0.073	0.042	144.00	30.35	2170.54 945.72 434.07 232.54	0.82 0.76 0.60 0.70	0.320 0.322 0.324 0.326	16.60 16.58 16.55 16.51
33	Fe ₇₂ Hf ₈ Nb ₂ B ₁₈ [72]	~10	863 873 883	0.694 0.702 0.71	0.079	0.046	112.65	38.33	29436.88 18825.39 13950.14	0.78 0.93 0.83	0.358 0.359 0.360	15.48 15.44 15.39
Geological Materials & Ice												
34	Limestone [73]	4.2	973 1073 1173	0.604 0.666 0.728	0.073	0.042	152.98	12.89	13761.67 7889.49 3085.08	0.57 0.68 0.38	0.321 0.326 0.329	15.80 15.76 15.64
35	Anorthite-Diopside, dry [74]	3.1	1323 1373 1423	0.726 0.753 0.781	0.062	0.036	165.03	24.03	10762.47 8091.02 5178.29	1.12 1.14 1.06	0.321 0.322 0.322	15.89 15.79 15.68
36	Anorthite-Diopside, wet [74]	3.1	1223 1273 1323 1373	0.671 0.699 0.726 0.753	0.062	0.036	142.75	24.36	22018.63 16955.57 8568.33 7164.01	1.27 1.03 1.04 1.14	0.319 0.320 0.321 0.322	16.03 15.97 15.89 15.79
37	Ice [75]	10	220 240 250	0.806 0.879 0.916	0.115	0.067	117.39	6.65	6227.10 4505.52 3443.24	0.79 0.59 0.49	0.289 0.292 0.293	16.67 16.68 16.69
38	Ice [76]	1700	241 256 263	0.883 0.938 0.963	0.1172	0.0677	144.4142	6.56806	1391.93 1318.684 1245.437	1.437 1.31 1.288	0.414 0.424 0.435	2.3 2.2 2.7
Nanostructured Materials												
39	Si ₃ N ₄ [77]	0.07	1723 1773 1823 1873	0.793 0.816 0.839 0.862	0.085	0.049	140.67	54.29	5945.67 3285.78 2724.33 2618.49	1.05 0.79 0.92 0.71	0.278 0.276 0.274 0.272	17.36 17.25 17.11 16.95
40	Zirconia- Alumina [78]	0.06	1623 1673 1723	0.584 0.602 0.62	0.069	0.040	109.91	27.94	2321.02 395.83 50.38	0.94 0.89 0.96	0.280 0.279 0.277	17.55 17.47 17.37

(continued on next page)

Table 1 (continued)

Sl. No.	System	Grain size (μm)	T (K)	T_{hom}	γ_0	ε_0	Q/T_m (J/Mol K)	G/T_m (MPa/K)	τ_0/T_m (Pa/K)	γ_B (J/m ²)	a -	N_a -
41	Zirconia- Spinel [78]	0.05	1763	0.635	0.075	0.043	120.89	21.78	28.77	0.93	0.276	17.28
			1573	0.574					5912.42	1.62	0.281	17.62
			1623	0.592					4452.56	1.68	0.280	17.56
			1673	0.61					2591.22	1.41	0.279	17.48
			1723	0.629					1569.35	1.51	0.277	17.38

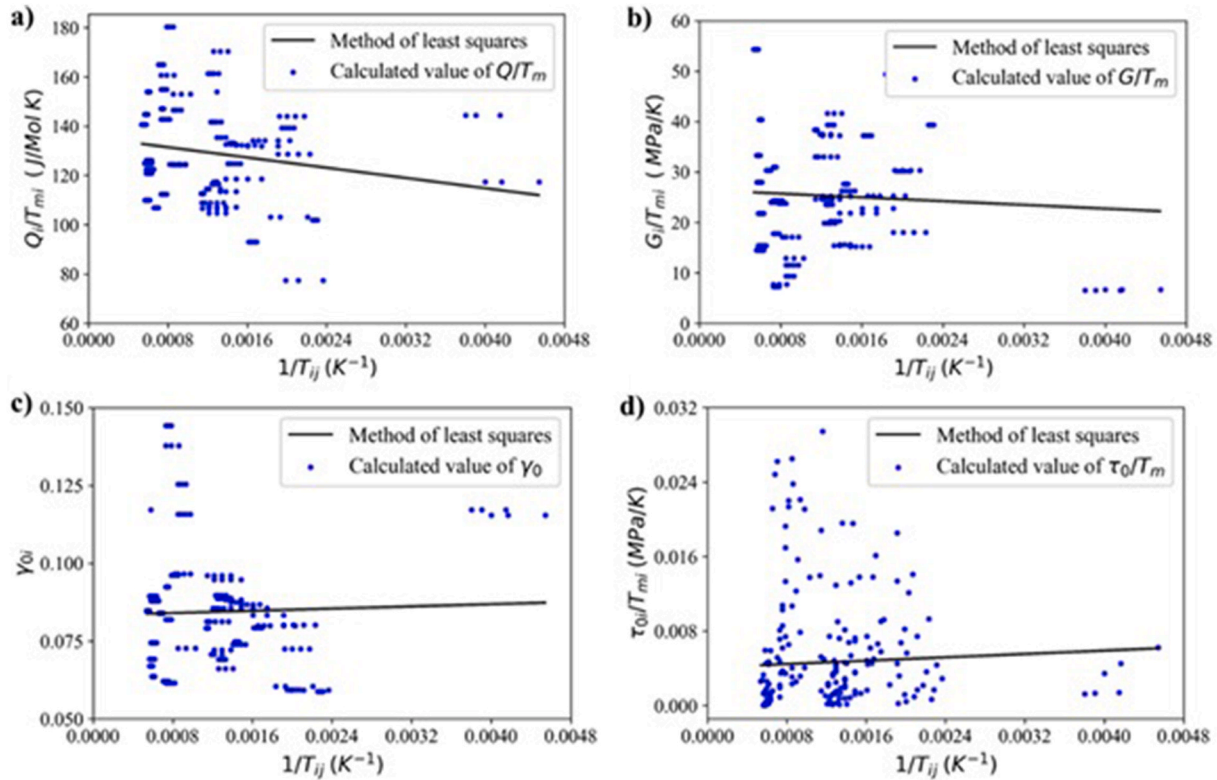


Fig. 1. a) Variation of Q/T_m with temperature, b) Variation of G/T_m with temperature, c) Variation of γ_0 with temperature, d) Variation of τ_0/T_m with temperature. Here “i” and “j” in the term stands for the value corresponding to the “ith” material condition and the value at temperature for every material condition “i”.

strain rates with the experimental values and the corresponding tolerance values are given in Table 3. The RMSE value in prediction is 0.0086 and the percentage of prediction that has a tolerance value above 10 is just 13 %. Evidently, the fits under all the thermomechanical conditions are very good and there was no need to omit any of the experimental data. Evidently the relatively higher error in prediction in a small number of observations is due to experimental scatter.

Based on the above analysis, the following can be briefly stated.

1. The validity of the model is claimed after examining 146 material conditions that exhibit isotropic steady state structural superplastic deformation over wide strain rate and temperature ranges and hence the inference can be regarded as broad-based.
2. If one notes the idea commonly invoked in statistics that “fit is least good at the periphery and widely scattering values at the periphery can be neglected”, it is clear that the consistency of the model constants holds good with a “tolerance” (the predicted/ observed values) less than ± 10 , the value stipulated for such predictions to be accepted as proof. The tolerance for only 15% of the 146 systems analysis is slightly over 10.
3. The predictions of those systems not used at all to derive the model constants are accurate with a tolerance less than 10 for all systems without exception.

4.3. Some observations on the predictions of the physical model

The physical picture presented in the model and its development thereof led to a few correct predictions when they were compared with the experimental findings. The details of the findings have been presented in the earlier sections. The predictions are stated below.

1. The stress-strain rate relation (the full curve is sigmoidal in shape, with the optimal/“pure” region of isotropic steady state structural superplastic flow of a parabolic shape) is predicted correctly. The dependences of strain rate on the applied stress, the grain size, and the test temperature are predicted correctly.
2. The variation of the instantaneous slope of the isothermal, iso-structural stress-strain rate plot, known as the strain rate sensitivity index, m , as a function of test temperature, grain size, and stress/strain rate is also predicted correctly.
3. The model can be applied to any material of any crystal structure and material class, prepared in appropriate microstructural conditions by prior treatments. This statement is validated in the paper, among others, by using the model to understand the isotropic, steady state structural superplastic deformation in different classes of materials with different crystal structures.

Table 2

Prediction accuracy for material conditions used for the analysis.

Sl. No.	System	Grain size (μm)	Temp (K)	$\dot{\epsilon}_{\text{exp}}$ (s^{-1})	$\dot{\epsilon}_{\text{pred}}$ (s^{-1})	Tolerance
1	Zn 22Al [41]	2.5	423	0.002092	0.0002195	9.53
			423	0.00497	0.000504	9.86
			423	0.005082	0.000544	9.34
			423	0.019677	0.00129	15.26
			473	0.005554	0.0038	1.46
			473	0.011298	0.003319	3.40
			473	0.023501	0.006618	3.55
			473	0.024248	0.007562	3.21
			503	0.021032	0.001765	11.92
			503	0.020571	0.004326	4.75
			503	0.045552	0.017464	2.61
			1273	0.000026	0.000519	19.96
			1273	0.000062	0.001049	16.92
			1323	0.000026	0.000212	8.15
			1323	0.000065	0.00065	10.0
17	Ti 43Al [56]	5	1323	0.00008	0.00080	10.0
			1323	0.000136	0.001329	9.77
			1323	0.000153	0.001474	9.63
			1323	0.000272	0.002369	8.71
			1373	0.000063	0.000299	4.75
			1373	0.000072	0.000452	6.28
			1373	0.00012	0.00111	9.25
			1373	0.000142	0.001373	9.67
			1373	0.000185	0.00186	10.05
			1273	0.000016	0.000015	1.07
			1273	0.00005	0.000016	3.07
			1273	0.000168	0.000019	8.75
			1273	0.000381	0.00021	1.83
			1323	0.000018	0.000049	2.72
			1323	0.000052	0.000053	1.02
22	ZrO ₂ [61]	0.07	1323	0.000172	0.000063	2.72
			1323	0.000505	0.000072	7.01
			1323	0.000916	0.000078	11.81
			1373	0.000018	0.000081	4.50
			1373	0.000054	0.000135	2.50
			1373	0.000172	0.000181	1.05
			1373	0.000514	0.000222	2.32
			1373	0.001008	0.000244	4.13
			483	0.002778	0.006871	2.47
			483	0.008454	0.010971	1.30
			493	0.008397	0.00447	1.88
			493	0.027586	0.008526	3.24
			503	0.029408	0.003739	7.87
			503	0.085413	0.008097	10.55
			513	0.084126	0.003962	21.23
31	La ₅₅ Al ₂₅ Ni ₂₀ [70]	~10	513	0.183592	0.008413	21.82
			973	0.000017	0.000162	9.53
			973	0.000017	0.000149	8.76
			973	0.000009	0.0000149	1.66
			1073	0.000045	0.00044	9.78
			1073	0.000036	0.000411	11.41
			1073	0.000031	0.000294	9.48
			1073	0.000024	0.000238	9.92
			1073	0.000017	0.0000194	1.14
			1073	0.000009	0.0000192	2.13
			1173	0.000147	0.000516	3.51
			1173	0.000123	0.000327	2.66
			1173	0.000016	0.000067	4.19
			1173	0.000016	0.000058	3.63
			1623	0.000543	0.000624	1.15
34	Limestone [73]	4.2	1623	0.001044	0.000142	7.33
			1673	0.0017	0.001685	1.01
			1673	0.003655	0.000473	7.73
			1723	0.009128	0.003788	2.41
			1763	0.009734	0.009127	1.07
40	Zirconia-Alumina [78]	0.06	1623	0.001044	0.000142	7.33
			1673	0.0017	0.001685	1.01
			1673	0.003655	0.000473	7.73
			1723	0.009128	0.003788	2.41
			1763	0.009734	0.009127	1.07

4. The model predicts that grain boundary sliding develops to a mesoscopic scale (of the order of a grain boundary length or more) and plane interfaces get formed. Such plane interfaces have been experimentally observed in different materials.

Table 3

Prediction accuracy for material conditions not used for the analysis.

Sl. No.	System	Grain size (μm)	Temp (K)	$\dot{\epsilon}_{\text{exp}}$ (s^{-1})	$\dot{\epsilon}_{\text{pred}}$ (s^{-1})	Tolerance
1	Zn 22Al [79]	3.7	450	5.98E-05	7.33E-06	8.17
			450	0.000109	4.45E-05	2.45
			450	0.000158	9.75E-05	1.62
			450	0.000271	0.000157	1.72
			450	0.002739	0.000657	4.17
			450	0.026074	0.002136	12.21
			500	5.37E-05	0.00016	2.98
			500	0.000104	9.80E-05	1.06
			500	0.000266	3.63E-05	7.33
			500	0.000456	0.000252	1.81
			500	0.00261	0.000613	4.26
			500	0.026284	0.002768	9.49
			525	9.30E-05	0.00058	6.23
			525	0.000156	0.000543	3.49
			525	0.000268	0.000399	1.49
			525	0.000707	0.000113	6.28
			525	0.001759	0.000467	3.77
			525	0.00255	0.000641	3.98
			525	0.026413	0.004498	5.87
			703	0.00078	0.001526	1.96
			703	0.001583	0.003466	2.19
			703	0.00243	0.004716	1.94
			703	0.003273	0.006607	2.02
			703	0.006046	0.010514	1.74
			703	0.007008	0.012362	1.76
			703	0.010877	0.016222	1.49
2	Al 6Cu 0.4Zr [80]	2.7	703	0.011831	0.016393	1.39
			703	0.019004	0.02114	1.11
			703	0.019573	0.022426	1.15
			703	0.032497	0.027968	1.16
			703	0.037003	0.031288	1.18
			723	0.001349	0.002103	1.56
			723	0.001413	0.002809	1.99
			723	0.002715	0.006246	2.30
			723	0.003016	0.008	2.65
			723	0.006291	0.014237	2.26
			723	0.006899	0.015936	2.31
			723	0.007794	0.017678	2.27
			743	0.002443	0.006099	2.50
			743	0.00309	0.00902	2.92
			743	0.006073	0.017928	2.95
			743	0.00691	0.0217	3.14
			743	0.01148	0.031711	2.76
			743	0.019337	0.044994	2.33
			763	0.002153	0.000219	9.83
			763	0.003167	0.007692	2.43
			763	0.005756	0.018302	3.18
			763	0.006928	0.025698	3.71
			763	0.011698	0.040135	3.43
			763	0.013856	0.044565	3.22
3	Pb-Sn eutectic [81]	7.5	311	1.93E-05	7.38E-05	3.83
			311	9.68E-05	0.000831	8.59
			328	9.33E-05	2.10E-05	4.45
			328	0.0002	0.000517	2.59
			348	0.00019	0.002004	0.09
			348	0.000336	0.000874	2.60
			348	0.000775	0.000201	3.85
			359	0.0002	0.000879	4.40
			359	0.000375	0.001184	3.15
			573	0.001415	0.000106	13.38
4	B-SiC _w / ZK60 Composite [82]	2.3	573	0.014222	0.000791	17.98
			613	0.001429	0.000344	4.15
			613	0.014222	0.000664	21.43
			613	0.028692	0.00091	31.54
			643	0.001415	0.000881	1.61
			673	0.001429	0.001877	1.31
			673	0.003618	0.002436	1.49
5	pure Ice [83]	312.5	723	0.001415	0.005506	3.89
			723	0.00534	0.006755	1.26
			253	1.62E-06	4.28E-07	3.78
			253	4.28E-06	8.56E-07	5.00
			253	9.04E-06	1.05E-06	8.61

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Table 3 (continued)

Sl. No.	System	Grain size (μm)	Temp (K)	$\dot{\epsilon}_{\text{exp}}$ (s^{-1})	$\dot{\epsilon}_{\text{pred}}$ (s^{-1})	Tolerance
6	PureIce+10% CaF ₂ [83]	312.5	253	1.80E-05	1.25E-06	14.42
			253	2.34E-05	1.32E-06	17.73
			263	2.63E-06	5.26E-07	5.01
			263	4.54E-06	6.63E-07	6.85
			263	9.35E-06	5.54E-07	16.88
			263	1.85E-05	6.63E-07	27.97
7	Mg 3Zn 0.5Y 1.5Zr [84]	5	253	1.78E-06	3.70E-07	4.82
			623	0.000351	0.000221	1.59
			623	0.000871	0.000486	1.79
			673	0.000351	0.000198	1.77
			673	0.000895	0.001085	1.21
			673	0.001861	0.002078	1.12
			698	0.001805	0.00211	1.17
			723	0.001803	0.001999	1.11
			1473	2.98E-06	6.95E-05	23.32
			1473	5.68E-06	9.09E-05	15.99
8	Olivine, Fo90 [85]	16	1473	1.20E-05	0.000116	9.72
			1473	2.42E-05	0.000148	6.09
			1473	5.20E-05	0.000189	3.64
			1523	4.43E-06	8.28E-05	18.66
			1523	5.44E-06	0.00011	20.30
			1523	9.49E-06	0.000141	14.90
			1523	1.73E-05	0.000182	10.51
			1523	3.55E-05	0.000235	6.61
			1673	0.000171	0.00289	16.87
			1723	0.001144	0.001501	1.31
9	Si ₂ O ₁₃ [86]	0.4	1773	0.006277	0.006802	1.08
			1773	0.010857	0.029341	2.70

- The dependence of the length of these mesoscopic scale plane interface and of N_a , the number of grain boundaries that align to form a plane interface, on grain size and temperature have been predicted correctly. The assumption of the model that N_a is independent of stress/strain rate has also been confirmed using the experimental results.
- Mechanisms for the removal of triple junctions and grain boundary obstacles have been correctly proposed in the model, as reported by Ikuhara et al. [18,19] and Wang et al. [21].
- Atomistic mechanisms for grain boundary sliding and grain boundary migration – atom shuffling parallel and perpendicular to the grain boundary respectively – were proposed correctly more than 25 years before experimental evidence [18,19,21] was obtained to validate these proposals.
- The prediction that the mesoscopic threshold stress will fall off to zero when the grain size reaches ~ 2.5 nm is yet to be verified. But it should be noted that so far it has not been possible to produce a material with a grain size less than ~ 2.5 nm.

As was pointed out in an earlier work [22], the entire analysis takes recourse to the mean field theory, which is an approximation. Based on this analysis, a case is made that mean field theoretical methods can also be used to understand other phenomena which display material class and crystal structure independence.

Alternatively, one can continue to have recourse to crystallography-based modelling. But it is safe to note that The CSL (Coincident Site Lattice) models and their adaptations/variants have been discredited in favour of Gleiter's structural unit model. However, in more recent times the crystallographic description has been revived by confining it to highly localised regions in the grain boundary, e.g., the concept of "disconnection" with its Burgers vector and obstacle height – to linearise the problem. Preservation of these disconnections during deformation does not seem to be realistic (see the papers of Ikuhara et al. [18,19] cited earlier) and so "complexion formation" has been proposed as an additional concept. Description of dislocation motion within such complexions and their glide along the grain boundaries of different

misorientations and climb, where necessary, constitutes a formidable theoretical problem, which is yet to be solved. In addition, in this approach superplastic deformation in every material will have to be considered separately.

It is often missed that problems like isotropic steady state structural superplastic deformation pertain to conditions far away from equilibrium where nonlinearity is the norm. Under such conditions pattern development and self-organisation prevail, in addition to non-linearity. The use of the mean field theory helps us simplify this formidable problem. In the opinion of the present authors, Gleiter's observation (personal communication) that the concept of "disconnection" seems to be an attempt at linearising a highly non-linear situation by coming up with a concept not much different from the discredited CSL concept to use it in a very small region in the grain boundary should be considered seriously.

But it is fully realised by the authors that in this analysis exact values are replaced by their mean values. In fact, in one of our earlier papers [16] we pointed out that because of the innumerable grain boundary misorientations and the differences in the atomic arrangements and the nature of the chemical bonds in each one of them, strictly speaking, it is appropriate to analyse the problem using a stochastic model. However, we preferred the mean value approach as it leads to a deterministic model, in addition to simplifying the physical description. It is inevitable that as the complexity of problems increase uncertainties also increase and this leads to inexactness. In our opinion, Einstein's comment, "As far as the laws of mathematics refer to reality, they are not certain; and so far as they are certain, they do not refer to reality" is relevant in this context.

5. Conclusions

A physical theory for isotropic steady state structural superplastic deformation, which is applicable to different classes of materials and different material conditions on a common basis has been analysed to obtain the model constants/parameters, Q , G , γ_B , γ_0 , ϵ_0 , τ_0 , N_a and a as functions of temperature and melting point of the material concerned (where necessary). The expressions for these parameters were obtained as mean values for 467 material conditions. A high degree of accuracy was obtained in the predictions for strain rates between 10^{-6} and 10^{-2} s^{-1} . The resultant percentage of predictions coming above a tolerance value of 10 could be reduced drastically if one omitted the strain rates at the periphery using the well-known statistical notion that always a fit is least good at the periphery. The accuracy of the developed model was further tested by making similar predictions for different material classes and initial conditions not used to obtain the material constants/parameters. The RMSE value and the percentage of predictions that come above a tolerance value of 10 are very few, which could be due to experimental scatter, and this suggests that the model that postulates that grain boundary sliding that develops to a mesoscopic scale controls the rate of isotropic steady state superplastic deformation is highly relevant to an understanding of this phenomenon in different classes of materials and crystal structures.

CRedit authorship contribution statement

K.R. Harisankar: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **K.A. Padmanabhan:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msea.2025.148175>.

Data availability

Data will be made available on request.

References

- [1] W.A. Backofen, I.R. Turner, D.H. Avery, Superplasticity in an Al-Zn alloy, *Trans. Am. Soc. Met.* 57 (1964) 980–990.
- [2] C.E. Pearson, The viscous properties of extruded eutectic alloys of lead-tin and bismuth-tin, *J. Inst. Met.* 54 (1934) 111–124.
- [3] K.A. Padmanabhan, G.J. Davies, *Superplasticity: Mechanical and Structural Aspects, Environmental Effects, Fundamentals and Applications*, Springer-Verlag, 1980.
- [4] O.A. Kaibyshev, *Superplasticity of Alloys, Intermetallics, and Ceramics*, Springer Berlin Heidelberg, 1992.
- [5] T.G. Nieh, J. Wadsworth, O.D. Sherby, *Superplasticity in Metals and Ceramics*, Cambridge University Press, Cambridge, 1997.
- [6] K.A. Padmanabhan, R.A. Vasin, F.U. Enikeev, *Superplastic Flow: Common Basis for a Ubiquitous Phenomenon*, Springer, New York, 2001.
- [7] K.A. Padmanabhan, S.B. Prabu, R.R. Mulyukov, A. Nazarov, R.M. Imaev, S. G. Chowdhury, *Superplastic Flow: Common Basis for a Ubiquitous Phenomenon*, Springer-Verlag, Berlin Heidelberg, 2018.
- [8] O. Engler, K.A. Padmanabhan, K. Lücke, A model for superplastic flow induced texture annihilation, *Model. Simulat. Mater. Sci. Eng.* 8 (2000) 477–490, <https://doi.org/10.1088/0965-0393/8/4/306>.
- [9] K.A. Babu, V.S. Sarma, C.N. Athreya, K.A. Padmanabhan, Experimental verification of grain boundary-sliding controlled steady state superplastic flow in both continually and statically recrystallizing Al alloys, *Mater. Sci. Eng. A* 657 (2016) 185–196, <https://doi.org/10.1016/j.msea.2016.01.054>.
- [10] K.A. Padmanabhan, Grain boundary sliding controlled flow and its relevance to superplasticity in metals, alloys, ceramics and intermetallics and strain-rate dependent flow in nanostructured Materials, *J. Mater. Sci.* 44 (2009) 2226–2238, <https://doi.org/10.1007/s10853-008-3076-1>.
- [11] K.A. Padmanabhan, A theory of structural superplasticity, *Mater. Sci. Eng.* 29 (1977) 1–18, [https://doi.org/10.1016/0025-5416\(77\)90140-9](https://doi.org/10.1016/0025-5416(77)90140-9).
- [12] K.A. Padmanabhan, J. Schlipf, Model for grain boundary sliding and its relevance to optimal structural superplasticity Part 1—theory, *Mater. Sci. Technol.* 12 (1996) 391–399, <https://doi.org/10.1179/026708396790165920>.
- [13] V.V. Astanin, S.N. Faizova, K.A. Padmanabhan, Model for grain boundary sliding and its relevance to optimal structural superplasticity Part 2 – evidence for cooperative grain/interphase boundary sliding and plane interface formation, *Mater. Sci. Technol.* 12 (1996) 489–494, <https://doi.org/10.1179/mst.1996.12.6.489>.
- [14] T.A. Venkatesh, S.S. Bhattacharya, K.A. Padmanabhan, J. Schlipf, Model for grain boundary sliding and its relevance to optimal structural superplasticity, *Mater. Sci. Technol.* 12 (1996) 635–643, <https://doi.org/10.1179/mst.1996.12.8.635>.
- [15] K.A. Padmanabhan, H. Gleiter, Optimal structural superplasticity in metals and ceramics of microcrystalline- and nanocrystalline-grain sizes, *Mater. Sci. Eng. A* 381 (2004) 28–38, <https://doi.org/10.1016/j.msea.2004.02.054>.
- [16] K.A. Padmanabhan, H. Gleiter, A mechanism for the deformation of disordered states of matter, *Cur. Opin. Sol. Sta Mater. Sci.* 16 (2012) 243–253, <https://doi.org/10.1016/j.cossms.2012.05.001>.
- [17] K.A. Padmanabhan, M. Basariya, A theory of steady state structural superplasticity in different classes of Materials: a Materials-agnostic analysis, *Mater. Sci. Eng. A* 744 (2019) 704–715, <https://doi.org/10.1016/j.msea.2018.12.086>.
- [18] J. Wei, B. Feng, R. Ishikawa, T. Yokoi, K. Matsunaga, N. Shibata, Y. Ikuhara, Direct imaging of atomistic grain boundary migration, *Nat. Mater.* 20 (2021) 951–955, <https://doi.org/10.1038/s41563-020-00879-z>.
- [19] J. Wei, B. Feng, E. Tochigi, N. Shibata, Y. Ikuhara, Direct imaging of the disconnection climb mediated point defects absorption by a grain boundary, *Nat. Commun.* 13 (2022) 1–6, <https://doi.org/10.1038/s41467-022-29162-2>.
- [20] P.M. Hazzledine, in: *Proc. 4th Risø Int. Symp. On Metallurgy And Materials Science, Roskilde, Denmark, September 1983*.
- [21] L. Wang, Y. Zhang, Z. Zeng, H. Zhou, J. He, P. Liu, M. Chen, J. Han, D.J. Srolovitz, J. Teng, Y. Guo, G. Yang, D. Kong, E. Ma, Y. Hu, B. Yin, X. Huang, Z. Zhang, T. Zhu, X. Han, Tracking the sliding of grain boundaries at the atomic scale, *Sci* 375 (2022) 1261–1265, <https://doi.org/10.1126/science.abm2612>.
- [22] K.A. Padmanabhan, S.V. Divinski, A mean field description of steady state isotropic structural superplastic deformation, *Mater. Sci. Eng. A* 908 (2024) 146713, <https://doi.org/10.1016/j.msea.2024.146713>.
- [23] S. Sripathi, K.A. Padmanabhan, On the experimental validation of a mesoscopic grain boundary sliding-controlled flow model for structural superplasticity, *J. Mater. Sci.* 49 (2014) 199–210, <https://doi.org/10.1007/s10853-013-7693-y>.
- [24] K.A. Padmanabhan, M. Basariya, A mesoscopic grain boundary sliding controlled flow model for superplasticity in intermetallics, *Int. J. Mater. Res.* 100 (2009) 1543–1551, <https://doi.org/10.3139/146.110218>.
- [25] S. Hawking, *On the Shoulders of Giants*, Running Press, London, 2003.
- [26] K.A. Padmanabhan, H. Gleiter, On the structure of grain/interphase boundaries and interfaces, *Beilstein J. Nanotechnol.* 5 (2014) 1603–1615, <https://doi.org/10.3762/bjnano.5.172>.
- [27] H. Hahn, K.A. Padmanabhan, A model for the deformation of nanocrystalline Materials, *Philos. Mag.* B 76 (1997) 559–571, <https://doi.org/10.1080/01418639708241122>.
- [28] K.A. Padmanabhan, G. Dinda, H. Hahn, H. Gleiter, Inverse Hall–Petch effect and grain boundary sliding controlled flow in nanocrystalline Materials, *Mater. Sci. Eng. A* 452–453 (2007) 462–468, <https://doi.org/10.1016/j.msea.2006.10.084>.
- [29] K.A. Padmanabhan, S. Sripathi, H. Hahn, H. Gleiter, Inverse Hall–Petch effect in quasi- and nanocrystalline Materials, *Mater. Lett.* 133 (2014) 151–154, <https://doi.org/10.1016/j.matlet.2014.06.153>.
- [30] J. Buenza, K.A. Padmanabhan, G. Wilde, On the applicability of a mesoscopic interface sliding controlled model for understanding superplastic flow in bulk metallic glasses, *Intermetallics* 60 (2015) 50–57, <https://doi.org/10.1016/j.intermet.2015.01.007>.
- [31] M.R. Basariya, N.K. Mukhopadhyay, S. Sripathi, K.A. Padmanabhan, Grain size softening effect in intermetallics, *J. Alloys Compd.* 673 (2016) 199–204, <https://doi.org/10.1016/j.jallcom.2016.02.258>.
- [32] V. Perevzentsev, V. Rybin, V. Chuvildeev, The theory of structural superplasticity—II. Accumulation of defects on the intergranular and interphase boundaries. Accommodation of the grain-boundary sliding. The upper bound of the superplastic strain rate, *Acta Metall. Mater.* 40 (1992) 895–905, [https://doi.org/10.1016/0956-7151\(92\)90066-N](https://doi.org/10.1016/0956-7151(92)90066-N).
- [33] K. A. Padmanabhan Gouthama, Transmission electron microscopic evidence for cavity nucleation during superplastic flow, *Scr. Mater.* 49 (2003) 761–766, [https://doi.org/10.1016/S1359-6462\(03\)00427-5](https://doi.org/10.1016/S1359-6462(03)00427-5).
- [34] N. Mara, A. Sergueeva, T. Mara, S. McFadden, A. Mukherjee, Superplasticity and cooperative grain boundary sliding in nanocrystalline Ni3Al, *Mater. Sci. Eng. A* 463 (2007) 238–244, <https://doi.org/10.1016/j.msea.2006.08.123>.
- [35] J. Markmann, P. Bunzel, H. Rosner, K. Liu, K.A. Padmanabhan, R. Birringer, H. Gleiter, J. Weissmuller, Microstructure evolution during rolling of inert-gas condensed palladium, *Scr. Mater.* 49 (2003) 637–644, [https://doi.org/10.1016/S1359-6462\(03\)00401-9](https://doi.org/10.1016/S1359-6462(03)00401-9).
- [36] T. Fliervoet, S. White, M. Drury, Evidence for dominant grain-boundary sliding deformation in greenschist- and amphibolite-grade polymetamorphic ultramylonites from the Redbank Deformed Zone, Central Australia, *J. Struct. Geol.* 9 (1997) 1495–1520, [https://doi.org/10.1016/S0191-8141\(97\)00076-X](https://doi.org/10.1016/S0191-8141(97)00076-X).
- [37] P. Haasen, *Physical Metallurgy*, Cambridge University Press, 1978.
- [38] A.S. Argon, Plastic deformation in metallic glasses, *Acta Metall.* 27 (1979) 47–58, [https://doi.org/10.1016/0001-6160\(79\)90055-5](https://doi.org/10.1016/0001-6160(79)90055-5).
- [39] J.H. Gittus, *Creep, Viscoelasticity and Creep Fracture in Solids*, Applied Sci. Publishers, London, 1975.
- [40] J.D. Eshelby, The determination of the elastic field of an ellipsoidal inclusion, and related problems, *Proc. Roy. Soc. Lond. A* 241 (1957) 376–396, <https://doi.org/10.1098/rspa.1957.0133>.
- [41] F.A. Mohamed, M.M. Ahmed, T. Langdon, Factors influencing ductility in the superplastic Zn-22%Al eutectoid, *Metall. Trans. A* 8 (1977) 933–938, <https://doi.org/10.1007/BF02661575>.
- [42] N. Kumar, K. Raman, D. Sastry, E. Little, Investigation of superplasticity by the impression creep technique, *J. Mater. Sci.* 25 (1990) 753–755, <https://doi.org/10.1007/BF00714105>.
- [43] K. Matsuki, Z. Horita, M. Tokizawa, Y. Murakami, Superplastic behaviour in nominally single-phase and two phase Al-Cu alloys, *Mater. Sci. Technol.* 13 (1979) 619–626, <https://doi.org/10.1179/msc.1979.13.11.619>.
- [44] T.G. Nieh, L.M. Hsiung, J. Wadsworth, R. Kaibyshev, High strain rate superplasticity in a continuously recrystallized Al-6%Mg-0.3%Sc alloy, *Acta Mater.* 46 (1998) 2789–2800, [https://doi.org/10.1016/S1359-6454\(97\)00452-7](https://doi.org/10.1016/S1359-6454(97)00452-7).
- [45] F. Musin, R. Kaibyshev, Y. Motohashi, G. Itoh, High strain rate superplasticity in a commercial Al-Mg-Sc alloy, *Scr. Mater.* 50 (2004) 511–516, <https://doi.org/10.1016/j.scriptamat.2003.10.021>.
- [46] S. Komura, T. Horita, M. Furukawa, M. Nemoto, T.G. Langdon, An evaluation of the flow behavior during high strain rate superplasticity in an Al-Mg-Sc alloy, *Metall. Mater. Trans. A* 32 (2001) 707–716, <https://doi.org/10.1007/s11661-001-0087-9>.
- [47] I. Charit, R.S. Mishra, Low temperature superplasticity in a friction-stir-processed ultrafine grained Al-Zn-Mg-Sc alloy, *Acta Mater.* 53 (2005) 4211–4223, <https://doi.org/10.1016/j.actamat.2005.05.021>.
- [48] R. Kaibyshev, E. Avtokratova, A. Apollonov, R.J.S.M. Davies, High strain rate superplasticity in an Al-Mg-Sc-Zr alloy subjected to simple thermomechanical processing, *Scr. Mater.* 54 (2006) 2119–2124, <https://doi.org/10.1016/j.scriptamat.2006.03.020>.
- [49] S. Fujino, N. Kuroishi, M. Yoshino, T. Mukai, Y. Okanda, K. Higashi, High-strain-rate superplastic behavior in a super-rapidly-solidified Al-Si system alloy, *Scr. Mater.* 37 (1997) 673–678, [https://doi.org/10.1016/S1359-6462\(97\)00153-X](https://doi.org/10.1016/S1359-6462(97)00153-X).

- [50] H. Watanabe, T. Mukai, K. Ishikawa, M. Mabuchi, K. Higashi, Realization of high-strain-rate superplasticity at low temperatures in a Mg–Zn–Zr alloy, *Mater. Sci. Eng. A* 307 (2001) 119–128, [https://doi.org/10.1016/S0921-5093\(00\)01974-2](https://doi.org/10.1016/S0921-5093(00)01974-2).
- [51] H. Watanabe, T. Mukai, K. Ishikawa, M. Mabuchi, K. Higashi, Superplasticity of a particle-strengthened WE43 magnesium alloy, *Mater. Trans.* 42 (2001) 157–162, <https://doi.org/10.2320/matertrans.42.157>.
- [52] H. Yan, R. Chen, E. Han, Superplasticity of Mg–5.8Zn–1Y–Zr alloy sheet fabricated by combination of extrusion and hot-rolling processes, *Sci. China Ser. E- Technol. Sci.* 52 (2009) 166–171, <https://doi.org/10.1007/s11431-008-0293-4>.
- [53] J. Babu, A. Dutta, Low temperature superplasticity through grain refinement in Ti–6Al–4V by a novel route of quench-roll-recrystallise, *J. Mater. Res. Technol.* 4 (2015) 348–352, <https://doi.org/10.1016/j.jmrt.2015.04.004>.
- [54] S. Shei, T.G. Langdon, The mechanical properties of a superplastic quasi-single phase copper alloy, *Acta Metall.* 26 (1978) 639–646, [https://doi.org/10.1016/0001-6160\(78\)90116-5](https://doi.org/10.1016/0001-6160(78)90116-5).
- [55] D.W. Livesey, N. Ridley, Superplastic deformation, cavitation, and fracture of microduplex Cu–Ni–Zn alloys, *Metall. Trans. A* 9 (1978) 519–526, <https://doi.org/10.1007/BF02646409>.
- [56] T.G. Nieh, J. Wadsworth, Microstructural characteristics and deformation properties in superplastic intermetallics, *Mater. Sci. Eng. A* 239–240 (1997) 88–96, [https://doi.org/10.1016/S0921-5093\(97\)00564-9](https://doi.org/10.1016/S0921-5093(97)00564-9).
- [57] G. Wegmann, R. Gerling, F.P. Schimansky, H. Clemens, A. Bartels, High-temperature mechanical properties of hot isostatically pressed and forged gamma titanium aluminide alloy powder, *Intermetallics* 10 (2002) 512–517, [https://doi.org/10.1016/S0966-9795\(02\)00026-2](https://doi.org/10.1016/S0966-9795(02)00026-2).
- [58] J. Sun, Y.H. He, J.S. Wu, Characterization of low-temperature superplasticity in a thermomechanically processed TiAl based alloy, *Mater. Sci. Eng. A* 329–331 (2002) 885–890, [https://doi.org/10.1016/S0921-5093\(01\)01646-X](https://doi.org/10.1016/S0921-5093(01)01646-X).
- [59] W. Kim, S. Hanada, T. Takasugi, Flow behavior and microstructure of Co3Ti intermetallic alloy during superplastic deformation, *Acta Mater.* 46 (1998) 3593–3604, [https://doi.org/10.1016/S1359-6454\(98\)00029-9](https://doi.org/10.1016/S1359-6454(98)00029-9).
- [60] T. Nieh, W.C. Oliver, Superplasticity of a nickel silicide, Ni3Si, *Scripta Metall.* 23 (1989) 851–854, [https://doi.org/10.1016/0036-9748\(89\)90258-5](https://doi.org/10.1016/0036-9748(89)90258-5).
- [61] M. Yoshida, Y. Shinoda, T. Akatsu, F. Wakai, Superplasticity-like deformation of nanocrystalline monoclinic zirconia at elevated temperatures, *J. Am. Ceram. Soc.* 87 (2004) 1122–1125, <https://doi.org/10.1111/j.1551-2916.2004.01122.x>.
- [62] F. Wakai, T. Nagano, Effects of solute ion and grain size on superplasticity of ZrO2 polycrystals, *J. Mater. Sci.* 26 (1991) 241–247, <https://doi.org/10.1007/BF00576058>.
- [63] A.A. Sharif, M.L. Mecartney, Superplasticity in cubic yttria-stabilized zirconia with intergranular silica, *Acta Mater.* 51 (2003) 1633–1639, [https://doi.org/10.1016/S1359-6454\(02\)00564-5](https://doi.org/10.1016/S1359-6454(02)00564-5).
- [64] T. Chen, F. Mohamed, M.L. Mecartney, Threshold stress superplastic behavior and dislocation activity in a three-phase alumina–zirconia–mullite composite, *Acta Mater.* 54 (2006) 4415–4426, <https://doi.org/10.1016/j.actamat.2006.05.002>.
- [65] R.P. Dillon, D.K. Kim, J.E. Trujillo, W.M. Kriven, M.L. Mecartney, Creep characteristics of alumina, nickel aluminate spinel, zirconia composites, *J. Mater. Res.* 23 (2008) 556–564, <https://doi.org/10.1557/JMR.2008.0071>.
- [66] T.D. Wang, J.C. Huang, High strain rate superplasticity in 6061 alloy with 1% SiO2 nano-particles, *Mater. Trans.* 42 (2001) 1781–1789, <https://doi.org/10.2320/matertrans.42.1781>.
- [67] Y. Nishida, I. Sigematsu, H. Arima, J.C. Kim, T. Ando, Superplasticity of SiC whisker reinforced 7075 composite processed by rotary-die equal-channel angular pressing, *J. Mater. Sci. Lett.* 21 (2002) 465–468, <https://doi.org/10.1023/A:1015378405808>.
- [68] Y. Kawamura, T. Shibata, A. Inoue, T. Masumoto, Superplastic deformation of Zr65Al10Ni10Cu15 metallic glass, *Scr. Mater.* 37 (1997) 431–436, [https://doi.org/10.1016/S1359-6462\(97\)00105-X](https://doi.org/10.1016/S1359-6462(97)00105-X).
- [69] M. Blety, P. Guyot, Y. Brechet, J. Blandin, J.L. Soubeyrou, Homogeneous deformation of Zr–Ti–Al–Cu–Ni bulk metallic glasses, *Intermetallics* 12 (2004) 1051–1055, <https://doi.org/10.1016/j.intermet.2004.04.019>.
- [70] Y. Kawamura, T. Nakamura, A. Inoue, T. Masumoto, High-strain-rate superplasticity due to Newtonian viscous flow in La55Al25Ni20 metallic glass, *Mater. Trans. JIM* 40 (1999) 794–803, <https://doi.org/10.2320/matertrans1989.40.794>.
- [71] Y. Saotome, T. Hatori, T. Zhang, A. Inoue, Superplastic micro/nano-formability of La60Al20Ni10Co5Cu5 amorphous alloy in supercooled liquid state, *Mater. Sci. Eng. A* 304–306 (2001) 316–320, [https://doi.org/10.1016/S0921-5093\(00\)01593-8](https://doi.org/10.1016/S0921-5093(00)01593-8).
- [72] Y. Kawamura, T. Itoi, T. Nakamura, A. Inoue, Superplasticity in Fe-based metallic glass with wide supercooled liquid region, *Mater. Sci. Eng. A* 304–306 (2001) 735–739, [https://doi.org/10.1016/S0921-5093\(00\)01600-2](https://doi.org/10.1016/S0921-5093(00)01600-2).
- [73] S.M. Schmid, Rheological evidence for changes in the deformation mechanism of Solenhofen limestone towards low stresses, *Tectonophysics* 31 (1976) T21–T28, [https://doi.org/10.1016/0040-1951\(76\)90160-8](https://doi.org/10.1016/0040-1951(76)90160-8).
- [74] A. Dimanov, E. Rybacki, R. Wirth, G. Dresen, Creep and strain-dependent microstructures of synthetic anorthite–diopside aggregates, *J. Struct. Geol.* 29 (2007) 1049–1069, <https://doi.org/10.1016/j.jsg.2007.02.010>.
- [75] W. Durham, L. Stern, S.H. Kirby, Rheology of ice I at low stress and elevated confining pressure, *J. Geophys. Res. Solid Earth* 106 (2001) 11031–11042, <https://doi.org/10.1029/2000JB900446>.
- [76] T.H. Jacka, The time and strain required for development of minimum strain rates in ice, *Cold Reg. Sci. Technol.* 8 (1984) 261–268, [https://doi.org/10.1016/0165-232X\(84\)90057-0](https://doi.org/10.1016/0165-232X(84)90057-0).
- [77] X. Xu, T. Nishimura, N. Hiroaki, R.J. Xie, Y. Yamamoto, H. Tanaka, Superplastic deformation of nano-sized silicon nitride ceramics, *Acta Mater.* 54 (2006) 255–262, <https://doi.org/10.1016/j.actamat.2005.09.005>.
- [78] X. Zhou, D.M. Hulbert, J.D. Kuntz, J.E. Garay, A.K. Mukherjee, Superplasticity of the nanostructured binary systems of zirconia–alumina–spinel ceramics by spark plasma sintering process, in: *Advances in Ceramic Matrix Composites X 155: Proceedings of the 106th Annual Meeting of, The American Ceramic Society, Indianapolis, Indiana, USA, 2004*, <https://doi.org/10.1002/9781118408353>.
- [79] A. Arieli, A. Yu, A. Mukherjee, Low stress and superplastic creep behavior of Zn–22 Pct Al eutectoid alloy, *Metall. Trans. A* 11 (1980) 181–191, <https://doi.org/10.1007/BF02700455>.
- [80] R. Bricknell, A. Bentley, The activation energy for superplastic flow in Al–6 Cu–0.4 Zr, *J. Mater. Sci.* 14 (1979) 2547–2554, <https://doi.org/10.1007/BF00610621>.
- [81] R. Cheese, B. Cantoor, Superplasticity in splat-quenched Pb Sn eutectic, *Mater. Sci. Eng. A* 45 (1980) 83–93, [https://doi.org/10.1016/0025-5416\(80\)90073-7](https://doi.org/10.1016/0025-5416(80)90073-7).
- [82] F. Yan, K. Wu, G. Wu, L. Lee, M. Zhao, Superplastic deformation behavior of a 19.7 vol. % β -SiC w/ZK60 composite, *Mater. Lett.* 57 (2003) 1992–1996, [https://doi.org/10.1016/S0167-577X\(02\)01118-7](https://doi.org/10.1016/S0167-577X(02)01118-7).
- [83] C.A. Middleton, P.M. Grindrod, P. Sammonds, The effect of rock particles and D2O replacement on the flow behaviour of ice, *Phil. Trans. A* 375 (2017) 20150349, <https://doi.org/10.1098/rsta.2015.0349>.
- [84] Y. Kim, D. Bae, Superplasticity in fine grained Mg–Zn–Y–Zr alloy containing quasicrystalline particles, *Mater. Trans.* 45 (2004) 3298–3303, <https://doi.org/10.2320/matertrans.45.3298>.
- [85] L. Hansen, M. Zimmermann, D. Kohlstedt, Grain boundary sliding in SanCarlos olivine: flow-law parameters and crystallographic-preferred orientation, *J. Geophysical Res.* 116 (2011) B08201, <https://doi.org/10.1029/2011JB008220>.
- [86] T. Chen, M.L. Mecartney, A high-strain-rate alumina-based ceramic composite, *J. Am. Ceram. Soc.* 88 (2005) 1004–1006, <https://doi.org/10.1111/j.1551-2916.2005.00190.x>.